VF Asia Footwear Laboratory Testing Manual

Test Method for Seam Stitch Tear- Upper, Lining & Webbing	Test Method #: VF- B19 REF # :TM-R- 037 Page 1 of 4
Date Issued: May,2019 Revision #: 1 Reason for revision: Rectify the unit of measure from kg to kgf for strength of material	Created by: Nila Babia Erazo

PURPOSE AND SCOPE: To measure breaking strength of stitched seams in shoe upper and lining materials. This test method can also be used to test webbing.

REFERENCES: SATRA TM 180

EQUIPMENT AND MATERIAL:

1. Tensile testing machine, meeting the requirements of ASTM D638
2. Clamping jaw,preferably rubber-faced
3. Clicker die capable of cutting test pieces 50 ± 2mm x 50 ± 2mm
4. A small hand knife or scissors for cutting test pieces
5. A sewing machine and accessories for making stitched seams
6. Fabric, leather, synthetic, webbing

SAMPLE PREPARATION:

Condition the test sample in a standard controlled environment @ 23.0± 2.0˚C temperature and 55 ±10 % Relative Humidity for at least 24 hours.

A. Upper and lining specimens:

1. Using the clicker die, scissors, or hand knife cut test pieces 50 ± 2mm x 50 ± 2mm from each material to be tested.
   - Three specimens shall be cut in the “A” direction, defined as the machine direction for synthetic materials and the direction parallel to the backbone for leather materials, three specimens shall be cut in the “B” direction, defined as 90° from the “A” direction.
2. Using the sewing machine stitch the test fabric to leather squares to produce three test seams for each direction of the test.
   - The leather squares should have a sufficiently high seam stitch tear strength such that failure first occurs in the test fabric.
   - Should have lapped seams using M-30, T-90, nylon thread and a 100 (16) needle.
   - The seams typically have 7.5 threads per inch. Alternate recommended stitching parameters can be found in SATRA Test Method PM 180, “Measurement of the Strength of Stitched Seams in Upper and Lining Materials,
3. Make cuts in each test specimen perpendicular to the seam from points 3mm from the seam to the edges of the material, to provide a test specimen having a central portion of width 25 ± 0.5mm and two edge pieces 12.5 ± 0.5mm, see Figure 1.

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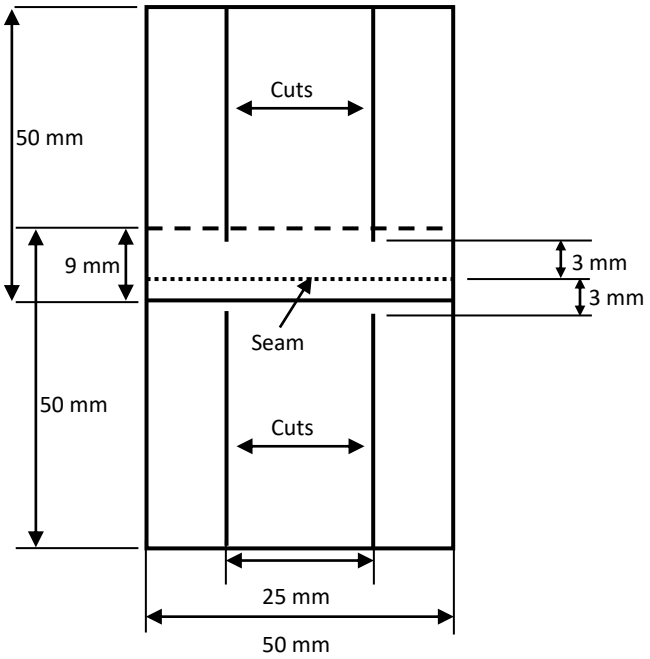


Figure 1. *Upper & Lining test specimen prepared by making up seam.*

B. For webbing that is **greater** than 25mm (1") wide, specimens should be prepared using the procedure outlined in section A.#3

C. For webbing that is **25mm (1") or less** in width, prepare webbing specimens as follows:

1. Using scissors or a hand knife cut test pieces 50 ± 2mm long from each webbing material to be tested.
2. Using the sewing machine stitch the webbing to leather squares to produce three test seams. The leather squares should have a sufficiently high seam stitch tear strength such that failure first occurs in the webbing. Timberland typically produces lapped seams using M-30, T-90, nylon thread and a 100 (16) needle. The seams typically have 7.5 threads per inch. Alternate recommended stitching parameters can be found in SATRA Test Method PM 180, "Measurement of the Strength of Stitched Seams in Upper and Lining Materials,

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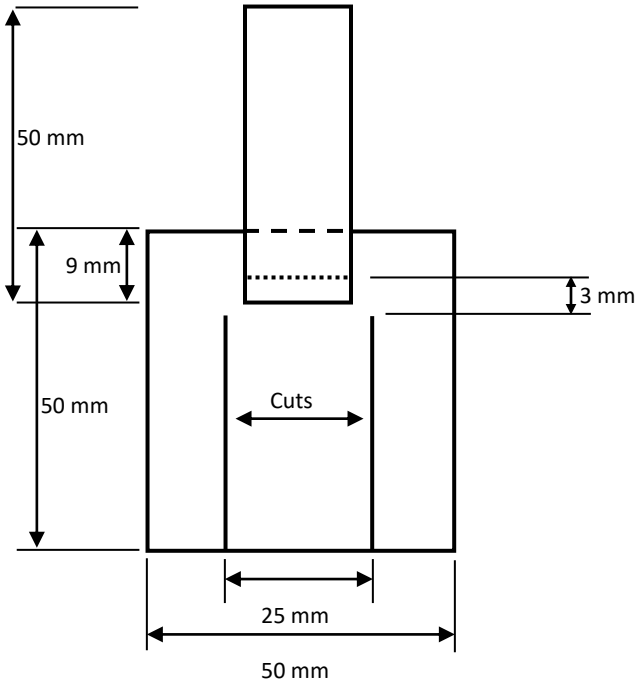


Figure 2. Webbing test specimen prepared by making up seam.

REPETITONS: Test three (3) samples

TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity. Samples are conditioned for at least 48 hours at the standard laboratory condition at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

1 For lining and upper materials, measure and record the width of the central portion of the test specimens in cm to the nearest cm. When a clicker die is used to prepare the specimens, the width can be assumed to be 2.5cm.
2. For webbing materials measure and record the width of the webbing in cm to the nearest cm.

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3. Adjust the distance between the jaws of the tensile testing machine so that it is approximately 30mm.
4. Clamp, midway between the jaws, the central portion of the test specimen between the jaws such that the seam is 15mm from each jaw and is parallel to the clamping edges. Do not clamp the 12.5mm edge strips, but allow them to hang free.
5. Operate the tensile testing machine at a speed of 100 ± 10 mm/min.
6. When the specimen fails, record the force in kgf at which failure occurs.

CALCULATION:

Breaking strength (kgf /cm) = B / W

B: breaking load at break in kgf

W: width of test piece in cm

REPORTS:

Seam Stitch Tear: kgf/cm

Also note the type of failure as:

1. Material failed at seam.
2. Thread pulled out from seam.
3. Thread failed.
4. Material failed, away from seam.